

General Course Information

Course name: Fundamentals of Agricultural Science

Course ID: MA615

Course Structure: 2 – 0 – 0 – 2

Course Type: Remedial course, Semester I for M.Sc. Agriculture Analytics

Credit : 2

Contact hours: 2

Course Objective: The course presumes that the participants have no exposure of subjects related to agriculture science and skills. The aim of this course would be to expose with basic information in agriculture e.g. agriculture economics, crop and soil science and agrometeorology. The level of information will be of undergraduate level.

Detailed content:

Theory

Sr. No.	Description	No. of Lectures
1	History of agriculture development in independent India and India's achievements in agriculture today, agricultural land holding and land reforms	1
2	Production trends of main food and cash crops; balance between food and cash crops	1
3	Factors affecting agriculture (Biotic and abiotic)	2
4	Soil as a medium affecting plant growth	1
5	Soil properties – physical, chemical and physicochemical	2
6	Soil water, soil organic matter, soil mineralogy	2
7	Soil plant relationship - Macronutrients	1
8	Soil classification and soil mapping	2
9	Field crops, cropping system, diseases of crops	2
10	Crop radiation capture and use efficiency	1
11	Spatial crop structure and its relation with yield	1
12	Simulation model for crop management	1
13	Agrometeorology and relevance to crop production	1
14	Solar radiation and crop production	1
15	Diurnal and seasonal variation in temperature, PET and importance to crop production	1
16	Introduction to precision agriculture	1
17	Weather forecasting	2
18	Impact of climate change on agriculture	1
19	Agriculture marketing and impact of climate change on agriculture value chain and supply chain	1
20	Climate smart agriculture	1
	Total	26

Evaluation criteria:

Evaluation is based on

- First in-semester exam and End semester exam (80%)
- Quizzes in the class (20%)

Learning outcome: After successful completion of the course, a student

- Will get exposure in agriculture economics and agriculture situation in India
- Will get exposure in crop and soil science
- Will get exposure in agrometeorology
- Will know about impact of climate change on agriculture

Learning and teaching activity:

- 2 x 1-hour lectures per week